

SREEDARSH T M

Remote sensing | Satellite Image Processing | Research Aspirant in GIS & RS



Research Portfolio- <https://sreedarsh.vercel.app>



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ABOUT ME

I have 4 years of practical experience in GIS and remote sensing, focusing on disaster management and forestry challenges like forest fires, landslides, floods, and coastal vulnerability I lean into vibe coding with AI platforms to quickly prototype tools, integrate smart workflows, and automate the boring, repetitive tasks. This lets me focus my energy on what matters most: exploring data and growing as a versatile researcher.

RESEARCH EXPERIENCE

July 2026 – Present

Geospatial Analyst | SatSure, Bengaluru, India

- Processing satellite imagery (multispectral, SAR) into actionable insights.
- Building models for crop monitoring, yield forecasting, and climate risk.
- Identifying and classifying features like crop types and land use.
- Creating high-quality reports, maps, and visual outputs for clients.
- Ensuring projects are delivered accurately and on time.

May 2026 – June 2026

Junior Executive (GIS) | Directorate of Census Operations, Kerala

- Contributing to the 16th Indian Census (Digital Census 2027)
- Collaborating with the MAP Division to maintain data integrity and spatial accuracy for multi-thematic census datasets.

Jan 2025 – June 2025

Project Intern | National Remote Sensing Centre (NRSC), ISRO, Hyderabad, India

- Project: "Burned Area Extraction using AI & ML (Deep Learning).
- Developed an automated, cloud-based data ingestion pipeline in Google Colab using the Earth Engine Python API to efficiently query, filter, and export multi-temporal satellite imagery (Sentinel-2 and Landsat 8/9).
- Engineered interactive GUI widgets to streamline the extraction of 453 specific 31*31 km geographic grids across India, accelerating the preprocessing workflow for downstream machine learning architectures.

ACADEMIC QUALIFICATIONS

DEGREE	UNIVERSITY	SPECIALIZATION / STREAM	YEAR
M. Tech	Bharathidasan University	Geoinformatics	2025
M. Sc	Mangalore University	Geography	2023
B. Sc	Kannur University	Geography Sub: Geology, Statistics	2021
VHSE	Kerala Board of VHSE	ABFS (Science) Agriculture, Biology, Chemistry, Physics	2018

RESEARCH INTERESTS

- Areas of Ongoing Learning / Current Learning Goals)
- Remote Sensing, GIS, Digital Image Processing of Satellite Data
 - Microwave Remote Sensing (NISAR, SAR, INSAR)
 - Python & Earth Engine, Machine Learning, and Deep learning
 - Disaster Management/Spatial Analysis

ACCEPTED PAPERS / CHAPTERS

- 1. SP. Dhanabalan, R. Jegankumar, V. Sindhuja, K. Prakash, **TM. Sreedarsh**. Prioritization of Flood Susceptibility zones using MCDM techniques, TOPSIS, and the CF Model in the Manjalar Watershed: A Part of the Vaigai Basin, South India. [DOI/PDF](#)

MANUSCRIPTS IN PREPARATION

- 2. **T.M. Sreedarsh** & Poojith Kumar D P. "Coastal Vulnerability Assessment of Kasaragod and Kannur using Geospatial Techniques." (*Book Chapter*)

ACADEMIC PROJECT

Burned Area Extraction using AI & Deep Learning | M. Tech Major Project

- o Applied deep learning algorithms to automate the detection of burned areas from satellite imagery.
- o Created a labeled training dataset for identifying burned areas across India to train deep learning models.
- o Gained hands-on experience in processing satellite imagery using Google Colab and the Earth Engine API.

Landslide Susceptibility Modeling using Machine Learning | M. Tech Mini Project

- o Conducted bivariate analysis and ML techniques to map landslide risks in Kodaikanal and its environs.

Coastal Vulnerability Assessment using Geospatial Techniques | M. Sc Major Project

- o Analyzed coastal changes and vulnerability indices for the Kasaragod and Kannur regions.

TECHNICAL SKILLS

Programming & Cloud	Python, Google Earth Engine (GEE) API
Development Tools	VS Code, Cursor, Google Colab, Git & GitHub
GIS Software	ArcGIS and ArcGIS Pro, (DSAS), QGIS, Google Earth Pro
DIP Software	Basic ENVI
Microsoft Office	Word, PowerPoint, Excel
Generative AI Tools.	ChatGPT, Gemini etc

INVITED TALKS & WORKSHOPS

Resource Person: Geospatial Data Processing with Python | Oct 2025

- **Host:** Department of Geography, Kannur University.
- Delivered a one-day hands-on workshop on "Geospatial Data Processing and Visualization using Earth Engine API & Google Colab" for the Postgraduate Diploma in Geo-Informatics.

CONFERENCE PRESENTATIONS

- **Site Suitability Analysis for Solid Waste Disposal in Kasaragod Taluk using GIS Techniques**
 - o 13th International Conference on Multi-disciplinary Approach for Sustainable Development (UGIT), Bangalore University | Dec 2024
- **Coastal Vulnerability Assessment of Kasaragod and Kannur using Geospatial Techniques**
 - o International Conference on Clean Water, Good Health, Sustainable Cities & Communities (CWGHSCC), Lovely Professional University, Punjab | Oct 2023

WORKSHOPS & ADVANCED TRAINING

- **High-End Workshop (Karyashala) on Hyperspectral Sensing**
 - o Topic: Agriculture and Water Resources Management.
 - o Organizer: Accelerate Vigyan Scheme (SERB Initiative).
- **GIAN Workshop on SWAT Modeling (Soil & Water Assessment Tool)**
 - o Topic: Impact of Landuse/Landcover and Climate Change on Water Resources.
 - o Host: NIT Trichy (5-Day Workshop).
- **AI for Space Application Training Program**
 - o Host: Hex-Star Universe | Feb 2025
 - o Completed 25 hours of intensive training on AI applications in Space Science and Technology.

SEMINAR / CONFERENCE / WORKSHOPS / WEBINAR OVERVIEW

IIRS ISRO E-CLASS	BHUVAN TRAINING	ESRI-MOOC	NASA ARSET	NPTEL	OTHERS
10	3	2	2	-	4

INTERNATIONAL		NATIONAL		WORKSHOP	OTHERS
Journals	Conference	Journals	Conference		Books, Chapters, Monographs, Manuals
-	2	-	-	2	1

Certificates Links 

REFERENCES

- I. **M.S.S. Praveen**, Scientist/Engineer - 'SE', FEG/RSAA, National Remote Sensing Centre (NRSC), Indian Space Research Organization (ISRO), Dept. Of Space, Govt. Of India, Hyderabad- -500 037 (T.S.). Mail somasatyapraveen_m@nrsr.gov.in
- II. **Dr. R. Jegankumar**, M.Sc., M.Tech., Ph.D., Professor and Head, Department of Geography, School of Earth Sciences, Bharathidasan University, Tiruchirappalli - 620024, Tamil Nadu, Ph: [9894748564](tel:9894748564), jegankumar@bdu.ac.in

DECLARATION

I declare that the information furnished above is true and correct to the best of my knowledge and belief.

Place: Kanhangad

Signature: 

CONTACT



Languages – Malayalam, English, Tamil



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